

Phase 1 Requirements:

- Setup evidence:

The screenshot displays the VS Code interface during a test run. The Explorer sidebar on the left shows the project structure, highlighting the `test_account.py` file under the `tests` directory. The Test Coverage tab provides a visual overview of the code's execution status, with green bars representing covered lines. The main editor pane shows the source code of `test_account.py`, which includes several test functions such as `test_positive_deposit()`, `test_negative_or_zero_reject(values)`, and `test_withdraw_insufficient_funds PASSED`. A terminal window at the bottom shows the command `pytest --cov-report=term-missing` being executed, resulting in a detailed coverage report.

Name	Stmts	Miss	Cover	Missing
<code>models_init_...py</code>	7	0	100%	
<code>modelsaccount.py</code>	58	20	72%	30, 34, 53-56, 68-62, 73, 80, 84, 94, 96, 102-103
<code>tests_init_...py</code>	0	0	100%	
<code>tests\test_account.py</code>	58	5	91%	27-30, 363
TOTAL	123	25	80%	

- Test Coverage setup PR: <https://github.com/tylervu-unlv/Group-1/pull/37>
- Test Coverage Issue: <https://github.com/tylervu-unlv/Group-1/issues/36>
- Link to your issue and its acceptance criteria.
 - Positive Deposit Issue: <https://github.com/tylervu-unlv/Group-1/issues/44>
 - Zero/Negative Deposit Issue: <https://github.com/tylervu-unlv/Group-1/issues/46>
- Link to your PR.
 - Positive Deposit PR: <https://github.com/tylervu-unlv/Group-1/pull/45>
 - Zero/Negative Deposit PR: <https://github.com/tylervu-unlv/Group-1/pull/47>
- Link to your coverage report or results.

```
platform win32 -- python 3.14.7, pytest-8.4.2, pluggy-1.6.0 -- C:\Users\Chris\groupProject\Group-1\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Chris\groupProject\Group-1\test_coverage_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 10 items

tests/test_account.py::test_account_role_assignment PASSED [ 10%]
tests/test_account.py::test_invalid_role_assignment PASSED [ 20%]
tests/test_account.py::test_invalid_email_format PASSED [ 30%]
tests/test_account.py::test_positive_deposit PASSED [ 40%]
tests/test_account.py::test_negative_or_zero_result[0] PASSED [ 50%]
tests/test_account.py::test_negative_or_zero_result[0.0] PASSED [ 60%]
tests/test_account.py::test_negative_or_zero_result[-100] PASSED [ 70%]
tests/test_account.py::test_negative_or_zero_result[-100.0] PASSED [ 80%]
tests/test_account.py::test_valid_withdrawal PASSED [ 90%]
tests/test_account.py::test_withdraw_insufficient_funds PASSED [100%]

===== tests coverage =====
coverage: platform win32, python 3.14.7-final-0

Name      Stmts  Miss  Cover   Missing
-----
models\__init__.py  7      0  100%
models\account.py   58     16   72%   30, 34, 53-56, 60-62, 73, 80, 84, 94, 98, 102-103
tests\__init__.py    0      0  100%
tests\test_account.py  58      5   91%   27-30, 163

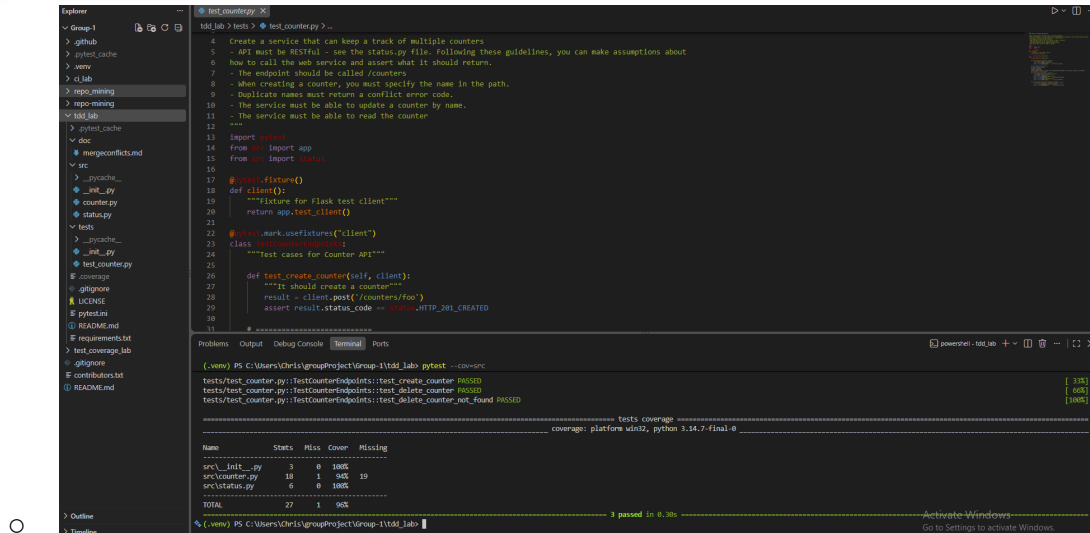
TOTAL          123     21   83%
```

- A brief explanation of the test and the behavior it verifies.
 - The test ensures that a positive amount increases the account balance and if it is a zero or negative value, then the deposit gets rejected, also checks decimal values.
- Link to the teammate's PR you reviewed and to your review comments.
 - PR review: <https://github.com/tylervu-unlv/Group-1/pull/58>
- A short summary of the feedback you received and how you addressed it.

- For my positive deposit PR, I had an extra quotation for line 8 and I simply removed it, fixing the requested change on the docstring.
 - For my zero/negative deposit, it was requested of me to fix a typo error in my acceptance criteria and to add an extra assert to my test to ensure no deposit went through for these invalid numbers. I addressed them by fixing the typo and adding that assert and notifying the reviewer of the changes
- Link to the approved PR or merge commit.
 - Shown above in the link to my PR.

Phase 2 Requirements:

- Setup evidence:



The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with files like `test_counter.py`, `counter.py`, and `status.py`. The code editor shows the content of `test_counter.py`, which includes a Flask app setup and several test functions. The bottom panel shows the output of running the tests, indicating that all tests passed.

```
4 Create a service that can keep a track of multiple counters
5 - API must be RESTful - see the status.py file. Following these guidelines, you can make assumptions about
6 how to call the web service and assert what it should return.
7 - The endpoint should be called /counters
8 - When creating a counter, you must specify the name in the path.
9 - Duplicate names must return a conflict error code.
10 - The service must be able to update a counter by name.
11 - The service must be able to read the counter.
12 ...
13 import pytest
14 from flask import app
15 from src import counter
16
17 @pytest.fixture()
18 def client():
19     """Fixture for flask test client"""
20     return app.test_client()
21
22 @pytest.mark.usefixtures("client")
23 class TestCounterAPI:
24     """Test cases for Counter API"""
25
26     def test_create_counter(self, client):
27         """It should create a counter"""
28         result = client.post("/counters/foo")
29         assert result.status_code == 201, "HTTP_201_CREATED"
30
31 # =====
```

Tests Coverage: coverage: platform win32, python 3.14.7-final-0

Name	Stats	Miss	Cover	Missing
src__init__.py	3	0	100%	
src\counter.py	18	1	94%	19
src\status.py	6	0	100%	
TOTAL	27	1	96%	

3 passed in 0.30s

- Tdd_lab setup Issue: <https://github.com/tylervu-unlv/Group-1/issues/52>
- Tdd_lab setup PR: <https://github.com/tylervu-unlv/Group-1/pull/53>
- Link to your issue and its acceptance criteria.
 - Incrementing a Counter and preventing update to non existent counter Issue: <https://github.com/tylervu-unlv/Group-1/issues/60>
- Link to your TDD PR and to the test case you wrote.
 - Incrementing a Counter and preventing update to non existent counter PR: <https://github.com/tylervu-unlv/Group-1/pull/61>
- A brief explanation of the feature being tested.
 - Test and ensure that when we encounter an existing counter, we increment it properly, and also tests for preventing updating a counter that does not exist and gives the proper status codes for both tests.
- Link to your RED/GREEN/REFACTOR evidence or test results.

- RED - As we see by just implementing tests, they initially fail

The screenshot shows a VS Code editor with a Python file named `test_counter.py`. The file contains three test functions: `test_create_counter`, `test_increment_counter`, and `test_non_existent_counter`. The `test_increment_counter` function is failing with an `AssertionError`. The output window shows the following error message:

```
tests/test_counter.py:47: AssertionError
===== tests coverage =====
coverage: platform win32, python 3.14.7-final-0

Name      Status  Miss  Cover  Missing
-----
src\init_.py      3      0  100%
src\counter.py    12      1   92%  19
src\status.py      6      0  100%
TOTAL          21      1   95%

short test summary info
FAILED tests/test_counter.py::TestCounterEndpoints::test_increment_counter - assert 405 == 200
FAILED tests/test_counter.py::TestCounterEndpoints::test_non_existent_counter - assert 405 == 404
===== 2 failed, 1 passed in 0.23s =====
```

- GREEN - After implementing the minimum code, the tests now pass

The screenshot shows the same VS Code editor with the `test_counter.py` file. The output window now shows that all tests have passed:

```
tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 33%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 66%]
tests/test_counter.py::TestCounterEndpoints::test_non_existent_counter PASSED [100%]

===== tests coverage =====
coverage: platform win32, python 3.14.7-final-0

Name      Status  Miss  Cover  Missing
-----
src\init_.py      3      0  100%
src\counter.py    18      1   94%  19
src\status.py      6      0  100%
TOTAL          27      1   90%

===== 3 passed in 0.10s =====
```

- REFACTOR - Was not needed because in the minimum code required to get the tests to pass we just use the `counter_exists()` method.

```
@app.route("/counters/<name>", methods=["PUT"])
def increment_counter(name):
    """increment a counter"""
    if not counter_exists(name):
        return jsonify({"error": f"Counter {name} does not exist"}), status.HTTP_404_NOT_FOUND
    COUNTERS[name] += 1
    return jsonify({name: COUNTERS[name]}), status.HTTP_200_OK
```

- A short summary of reviewer feedback and how you addressed it.
 - The reviewer stated that the tests worked fine however there were some merge conflicts with updates to the file that I did not have previously. I took note of this, and handled any merge conflicts that were present and ensured the code worked as intended. The reviewer also confirmed this was resolved as well.
- Link to the teammate's PR you reviewed and to your review comments.
 - PR review: <https://github.com/tylervu-unlv/Group-1/pull/72>
- A brief statement of the evidence you examined while reviewing (for example: test results, RED/GREEN evidence, coverage report).
 - The evidence that I garnered after reviewing my teammates PR showed the test results worked for their specific feature, and I ensured that the test failed first in the red phase and only passed after implementation of the minimum code required.
- Link to the approved PR or merge commit.
 - Shown above in the link to my PR.

Phase 3 Requirements:

Repository

- Link to your fork.
 - Fork: <https://github.com/Bigolbean7804/Group-1/tree/main>
- Evidence that your fork was synchronized with the upstream repository.

This branch is up to date with `tylervu-unlv/Group-1:main` . Contribute Sync fork

Bigolbean7804 Merge pull request `tylervu-unlv#72` from VegasBear/list-counters-V... 833d078 · 8 hours ago 116 Commits

<code>.github/workflows</code>	added workflow	15 hours ago
<code>ci_lab</code>	deleted <code>.DS_Store</code>	12 hours ago
<code>repo-mining</code>	Merge pull request <code>tylervu-unlv#30</code> from VegasBear/repo-mi...	2 weeks ago
<code>repo_mining</code>	Merge pull request <code>tylervu-unlv#32</code> from zombieg1rl/repo-m...	2 weeks ago
<code>tdd_lab</code>	Merge main and resolve conflicts	9 hours ago
<code>test_coverage_lab</code>	Merge pull request <code>tylervu-unlv#58</code> from VegasBear/test-pas...	8 hours ago
<code>.gitignore</code>	Updated <code>.gitignore</code> , based on CI lab starter repo	12 hours ago
<code>README.md</code>	Initial commit	3 weeks ago
<code>contributors.txt</code>	Resolve contributor list conflict	2 weeks ago

Branch is up to date with main upstream, can also check fork link above.

- Evidence that the `ci_lab` starter workflow ran successfully for you.

```
(www) PS C:\Users\Chris\groupProject\Group-1\ci_lab: pytest --cov=src --cov-report=term-missing
platform win32 -- python 3.14.7, pytest-8.4.2, pluggy-1.6.0 -- C:\Users\Chris\groupProject\Group-1\venv\Scripts\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\Chris\groupProject\Group-1\ci_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 22 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 45%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_duplicate_counter PASSED [ 91%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 136%]
tests/test_counter.py::TestCounterEndpoints::test_return_404_for_non_existent_counter PASSED [ 182%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 228%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_updating_non_existent_counter PASSED [ 274%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 319%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_deleting_non_existent_counter PASSED [ 365%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [ 409%]
tests/test_counter.py::TestCounterEndpoints::test_list_all_counters PASSED [ 455%]
tests/test_counter.py::TestCounterEndpoints::test_handle_invalid_http_methods PASSED [ 500%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_counters PASSED [ 546%]
tests/test_counter.py::TestCounterEndpoints::test_top_n_counters PASSED [ 592%]
tests/test_counter.py::TestCounterEndpoints::test_bottom_n_counters PASSED [ 638%]
tests/test_counter.py::TestCounterEndpoints::test_set_counter_to_value PASSED [ 684%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_negative_counter_values PASSED [ 729%]
tests/test_counter.py::TestCounterEndpoints::test_reset_single_counter PASSED [ 775%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_resetting_non_existent_counter PASSED [ 821%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_number_of_counters PASSED [ 867%]
tests/test_counter.py::TestCounterEndpoints::test_counters_greater_than_threshold PASSED [ 913%]
tests/test_counter.py::TestCounterEndpoints::test_counters_less_than_threshold PASSED [ 959%]
tests/test_counter.py::TestCounterEndpoints::test_validate_counter_name PASSED [ 1000%]

===== tests coverage =====
coverage: platform win32, python 3.14.7-final-0

Name      Stats    Miss  Cover   Missing
src\__init__.py    3      0  100%
src\counter.py    87      5   94%   71, 85, 96, 99-100
src\status.py     7      0  100%
TOTAL          97      5   95%

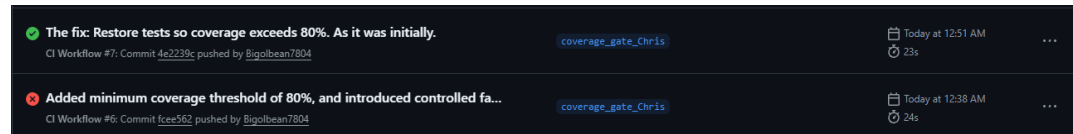
32 passed in 0.24s
```

Combined CI PR

- Link to your issue and its acceptance criteria.
 - Enforce a minimum coverage threshold of 80% Issue: <https://github.com/tylervu-unlv/Group-1/issues/71>

- Link to your PR.
 - Changed files are shown through commit history in PR link.
 - Enforce a minimum coverage threshold of 80% PR:
<https://github.com/tylervu-unlv/Group-1/pull/75>

- Link to the workflow excerpt or a screenshot of it.



-
- Link to the failing CI log and your root-cause summary.
 - Failing CI run:
<https://github.com/Bigolbean7804/Group-1/actions/runs/35320369813>
 - Root Cause Analysis: Removing 13 tests from the source tests_counter.py dropped our coverage of counter.py from 95% to 75.26%
- Link to the fix commit or PR diff.
 - Fix commit:
<https://github.com/tylervu-unlv/Group-1/pull/75/changes/4e2239c3dd05eccb2b7ab5d4ea3e4306954fed23>
- Link to the successful CI run.
- Successful CI run:
<https://github.com/Bigolbean7804/Group-1/actions/runs/35321485673>
- Link to the teammate's PR you reviewed and to your review comments.
 - PR review:
<https://github.com/tylervu-unlv/Group-1/pull/77#pullrequestreview-5251272182>

- A brief statement of the evidence you examined while reviewing (for example: CI logs, test results, workflow diff).
 - The matrix setup enhancement was implemented correctly. Controlled failure is clearly shown in the commit history. The CI logs support the root cause analysis of python 3.9 not being able to use the match keyword that was being used in line 132 of counter.py. The fix restores a successful CI run as indicated in the commit history.

Reflection

Write a short reflection on what you learned about continuous integration, automated testing, coverage and quality gates, CI failure diagnosis, technical peer review, and collaborative software development.

- Working through the CI lab taught me about the mechanics of how to use continuous integration. Implementing the coverage gate showed me how to implement a quality gate in practice. An automated check that fails a build when a codebase falls below a specific predefined threshold. Generally speaking, the exercise showed why CI matters for collaborative development. A shared automatic pipeline helps the team by giving a sort of “sanity check” to ensure a feature works with everyone.